ElderCareAI System Storyboard

This storyboard documents the main caregiver phone screens, Galaxy Watch6 screens, and administrator web screens. Each figure box identifies the screenshot to insert. The items and logic describe the current system at a general level; live screenshots should show the installed build and real demonstration records.

Module Name: Authentication Module

Screen No: 1

Screen Name: Phone sign in

Description: Caregivers start a Google sign-in session and continue to their saved profile or onboarding.

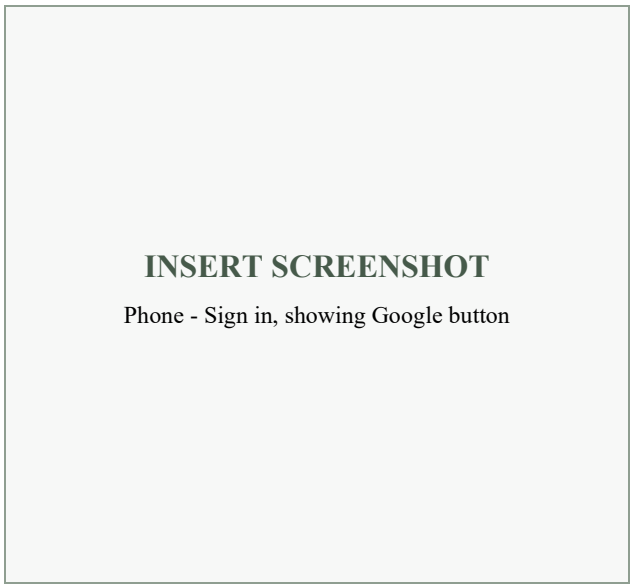


Figure 1: **Phone sign in**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
ElderCareAI logo	Image	No	N/A	N/A
Google sign-in	Button	Yes	OAuth action	N/A
Sign-in progress	Indicator	When loading	N/A	N/A
Sign-in error	Text	When failed	VARCHAR	N/A

Logic:

1. The app checks for an existing caregiver session when the sign-in screen opens.
    1.1. If a valid session exists, the app resolves the next onboarding step or opens Home.
    1.2. If no session exists, the Google sign-in control remains available.
2. The caregiver taps Google sign-in and completes the account flow.
    2.1. If the flow is cancelled or fails, the app shows a readable error and stays on sign-in.
    2.2. If authentication succeeds, the app restores the caregiver account context.
3. The app checks the caregiver, older-adult, watch, and location setup progress.
    3.1. If a required profile step is incomplete, the app opens that setup screen.
    3.2. If setup is complete, the app opens Home for the selected older adult.
4. On a later launch, the app checks the saved session again before displaying patient data.

Module Name: Onboarding Module

Screen No: 2

Screen Name: Caregiver setup

Description: The caregiver enters their own profile details before adding an older adult.

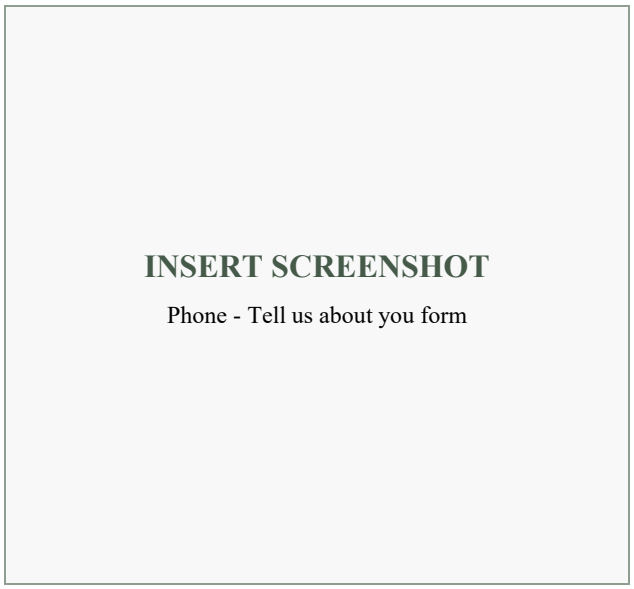


Figure 2: **Caregiver setup**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Caregiver photo	Image picker	No	FILE	N/A
Full name	Textbox	Yes	VARCHAR	120
Authenticated email	Read-only field	Yes	VARCHAR	N/A
Phone	Textbox	Yes	VARCHAR	N/A
Age	Numeric textbox	Yes	INTEGER	3 digits
Sex or gender	Choice chips	Yes	ENUM	N/A
Next or Save changes	Button	Yes	Action	N/A

Logic:

1. The app opens the caregiver form for the signed-in account.
    1.1. If a profile already exists, its editable details are loaded into the form.
    1.2. The authenticated email is displayed but cannot be changed here.
2. The caregiver may select a photo and enters full name, phone, age, and sex or gender.
3. The caregiver taps Next or Save changes.
    3.1. The app checks that required values are present and that phone and age values are valid.
        3.1.1. If a value is invalid, its field shows feedback and the entered data remains available.
        3.1.2. If fields are valid, the app submits the profile for the current account.
    3.2. If saving fails, the app reports the error without advancing.
4. If saving succeeds, onboarding advances to older-adult setup; edit mode returns with the revised profile.

Module Name: Onboarding Module

Screen No: 3

Screen Name: Older adult setup

Description: The caregiver records the selected older adult and emergency contact information.

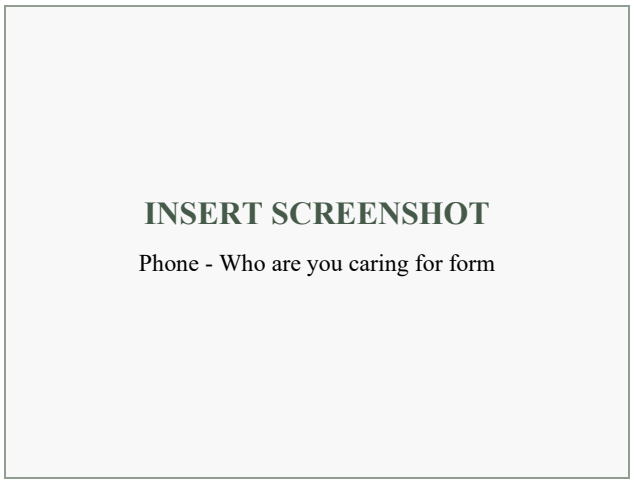


Figure 3: **Older adult setup**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Older adult photo	Image picker	No	FILE	N/A
Full name	Textbox	Yes	VARCHAR	120
Date of birth	Date picker	One of date or age	DATE	N/A
Age instead	Numeric textbox	One of date or age	INTEGER	3 digits
Gender	Choice chips	Yes	ENUM	N/A
Weight	Decimal textbox	Yes	NUMERIC	kg
Height	Decimal textbox	Yes	NUMERIC	cm
Blood type	Choice chips	Yes	ENUM	N/A
Medical conditions	Multiline textbox	No	TEXT	N/A
Current medications	Multiline textbox	No	TEXT	N/A
Emergency contact name	Textbox	Yes	VARCHAR	N/A
Emergency contact phone	Phone textbox	Yes	VARCHAR	N/A
Save profile	Button	Yes	Action	N/A

Logic:

1. The app opens the older-adult form for the current caregiver.
    1.1. In edit mode, it loads the existing patient's saved details.
2. The caregiver enters full name and either a date of birth or the alternate age.
    2.1. Selecting a birth date displays the calculated age and replaces the manual age entry.
3. The caregiver selects gender and blood type and enters weight and height in the shown units.
    3.1. Medical conditions and current medications may be added when known.
4. The caregiver enters the emergency contact name and phone number, then taps Save profile.
    4.1. The app checks required fields, age or birth date, physical values, and contact information.
        4.1.1. If validation fails, the affected fields show feedback and the form remains open.
        4.1.2. If validation passes, the app saves the profile under the current caregiver.
    4.2. If the database save fails, the app retains the form for correction or retry.
5. After a successful save, the app advances to Galaxy Watch6 setup.

Module Name: Wearable Connection Module

Screen No: 4

Screen Name: Galaxy Watch6 connection

Description: A watch is paired to the selected patient profile so measurements and location belong to the right person.

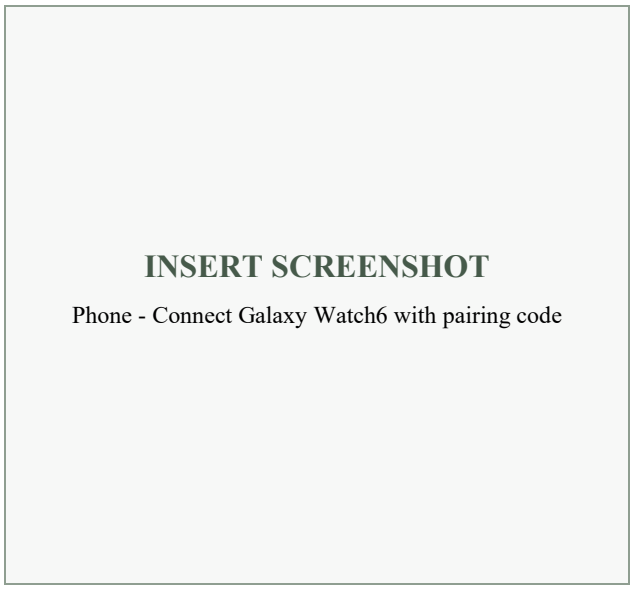


Figure 4: **Galaxy Watch6 connection**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Patient name	Text	Yes	VARCHAR	N/A
Connection status	Status text	Yes	ENUM	N/A
Pairing code	Code	When pairing	CHAR	8 digits
Connect or change watch	Button	Yes	Action	N/A
Set up later	Button	No	Action	N/A
Pair watch on watch	Watch button	When pairing	Action	N/A

Logic:

1. The app opens watch setup for the selected older adult.
    1.1. If a watch is already associated, the connection state is shown.
2. The caregiver chooses Connect or change watch to obtain a pairing code.
    2.1. The phone shows the code and the patient account it will connect to.
3. On the watch, the wearer opens ElderCareAI, enters the eight digits, and taps Pair watch.
    3.1. If the code is invalid, expired, or cannot reach the service, pairing fails visibly.
    3.2. If the code is accepted, the watch is associated with that patient and the phone shows Connected.
4. The caregiver may choose Set up later.
    4.1. The app can continue, but watch-dependent vitals and location remain unavailable until pairing.
5. A later watch change must be paired to the intended patient profile again.

Module Name: Galaxy Watch6 Module

Screen No: 5

Screen Name: Watch monitoring

Description: The watch starts a foreground monitoring session and displays the most recent available measurements.

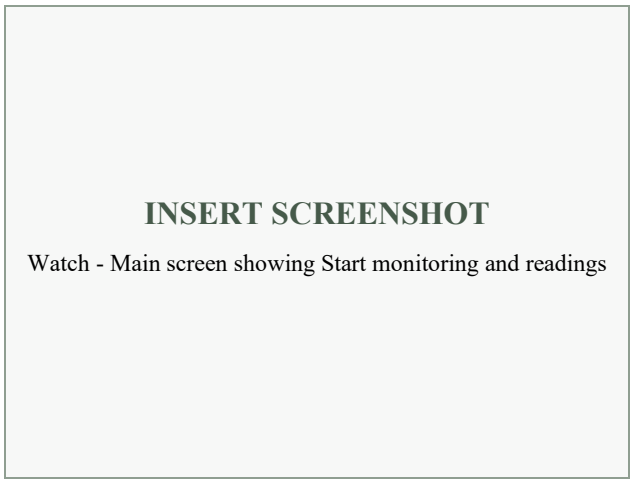


Figure 5: **Watch monitoring**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Heart rate value	Reading	When measured	NUMERIC	bpm
Measurement age	Text	When measured	DURATION	N/A
Start or stop monitoring	Button	Yes	Action	N/A
SOS	Button	Yes	Action	N/A
Medications	Button	Yes	Action	N/A
Skin temperature	Reading	When measured	NUMERIC	C
Stress estimate	Reading	When measured	NUMERIC	0-100
Monitored steps	Reading	When measured	INTEGER	steps
Blood oxygen	Reading	When measured	NUMERIC	%
Measure blood oxygen	Button	No	Action	N/A
Location status	Text	Yes	VARCHAR	N/A
Retry location	Button	No	Action	N/A
Sharing status	Text	Yes	VARCHAR	N/A

Logic:

1. The paired watch opens its dashboard and displays the latest available readings.
    1.1. A card without a valid reading shows its empty or waiting state.
2. The wearer taps Start monitoring.
    2.1. The watch requests required sensor and location permissions.
        2.1.1. If a permission is denied or a sensor is unavailable, the watch displays a status message.
        2.1.2. If access is granted, the foreground monitoring service requests measurements and location.
3. As accepted measurements arrive, the corresponding cards update with value, unit, and age.
    3.1. The watch queues accepted readings for upload when internet access is available.
    3.2. If upload fails, queued data is retried after connectivity returns.
4. If location has no fresh fix, the wearer may tap Retry location and check the status text.
5. The wearer may open Medications, the SOS confirmation, or a blood-oxygen spot check.
6. Tapping Stop monitoring ends the active session; saved readings finish uploading when possible.

Module Name: Galaxy Watch6 Module

Screen No: 6

Screen Name: Blood oxygen spot check

Description: The wearer can request a foreground SpO2 spot check; optional repeat checks require this screen to remain open.

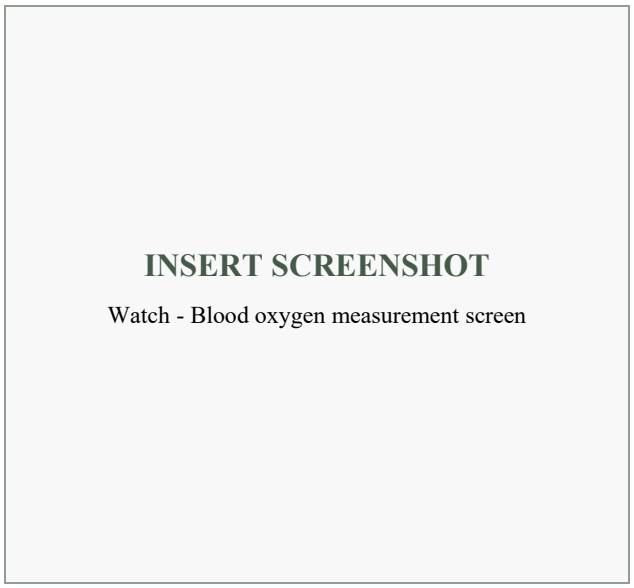


Figure 6: **Blood oxygen spot check**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Blood oxygen value	Reading	When measured	NUMERIC	%
Spot-check status	Text	Yes	VARCHAR	N/A
Measure blood oxygen	Button	Yes	Action	N/A
Auto checks every 5 min	Switch	No	BOOLEAN	N/A
Countdown or progress	Indicator	While measuring	INTEGER	seconds
Back to dashboard	Button	Yes	Action	N/A

Logic:

1. The wearer starts monitoring and opens the Blood oxygen screen.
    1.1. If monitoring is inactive, the screen asks the wearer to start it first.
2. The wearer keeps the watch snug, rests the arm, and taps Measure blood oxygen.
    2.1. The screen shows progress while the sensor attempts a spot measurement.
        2.1.1. If the sensor disconnects, times out, or returns no usable value, an error appears and retry remains possible.
        2.1.2. If a usable value arrives, the screen shows the SpO2 result and queues it for upload.
3. The wearer may enable Auto checks every 5 min.
    3.1. The next attempt occurs only while this screen remains active and the watch stays awake.
    3.2. Leaving the screen or stopping monitoring cancels the repeat checks.
4. Back to dashboard returns to the regular monitoring view.

Module Name: Caregiver Dashboard Module

Screen No: 7

Screen Name: Home dashboard

Description: The dashboard summarizes the selected older adult, latest saved readings, and current care tasks.

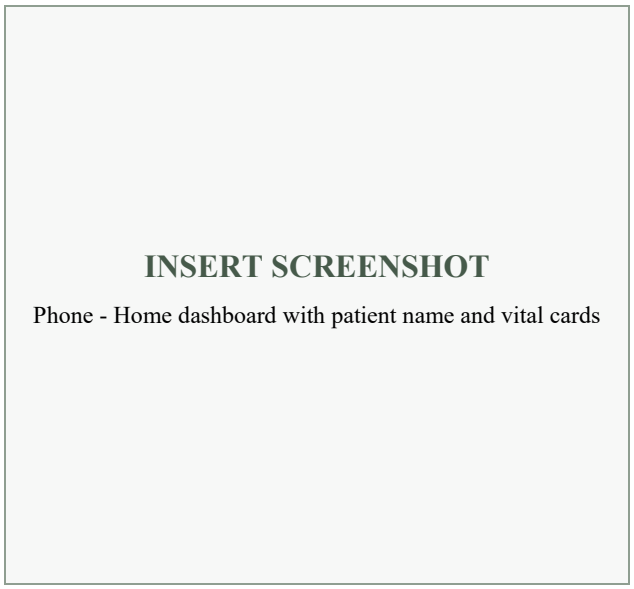


Figure 7: **Home dashboard**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Selected patient	Text	Yes	VARCHAR	N/A
Sync now	Button	No	Action	N/A
Heart rate card	Card	When recorded	NUMERIC	bpm
Blood oxygen card	Card	When recorded	NUMERIC	%
Sleep card	Card	When recorded	NUMERIC	hours
Steps card	Card	When recorded	INTEGER	steps
Skin temperature card	Card	When recorded	NUMERIC	C
Stress level card	Card	When recorded	NUMERIC	0-100
Latest care tasks	List	When saved	OBJECT	N/A
Doctor call	Button	When saved	Action	N/A

Logic:

1. Home loads the signed-in caregiver and selected older-adult profile.
    1.1. If the patient context is missing, the app directs the caregiver to the relevant setup step.
2. The app requests the patient's latest saved vitals and care tasks.
    2.1. Each available reading displays its value, unit, status text, and measurement time.
    2.2. Missing or stale data is labeled without substituting a zero or another patient's reading.
3. The caregiver may tap Sync now to request a fresh update.
    3.1. If refresh succeeds, changed cards and timestamps appear.
    3.2. If refresh fails, the last received readings remain visible with an error note.
4. Tapping a vital card opens its detail; tapping a care preview opens the related Care plan tab.
5. The doctor call action appears only when a doctor contact has been saved.

Module Name: Health Monitoring Module

Screen No: 8

Screen Name: Vitals

Description: The Health screen displays heart rate, blood oxygen, sleep, steps, wrist temperature, and the stress estimate when recorded.

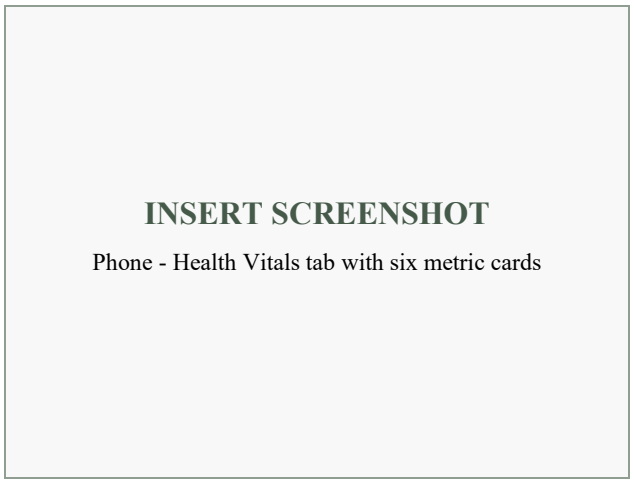


Figure 8: **Vitals**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Vitals tab	Tab	Yes	Action	N/A
Analytics tab	Tab	Yes	Action	N/A
AI Insights tab	Tab	Yes	Action	N/A
Predictions tab	Tab	Yes	Action	N/A
Anomalies tab	Tab	Yes	Action	N/A
Heart rate	Vital card	When recorded	NUMERIC	bpm
Blood oxygen	Vital card	When recorded	NUMERIC	%
Sleep	Vital card	When recorded	NUMERIC	hours
Monitored steps	Vital card	When recorded	INTEGER	steps
Skin temperature	Vital card	When recorded	NUMERIC	C
Stress level	Vital card	When recorded	NUMERIC	0-100
Status and time	Text	When recorded	VARCHAR	N/A

Logic:

1. The caregiver opens Health and selects Vitals.
2. The app loads the latest recorded values for the selected patient.
    2.1. It maps heart rate, blood oxygen, sleep, steps, skin temperature, and stress to separate cards.
    2.2. If a metric has no reading, that card displays a missing-data state.
3. For a recorded metric, the app shows the value, unit, measurement time, and readable status label.
    3.1. The label appears with color so meaning does not depend on color alone.
4. When a newer accepted reading arrives, the relevant card and its status update.
5. Tapping a card opens its detail for that metric.
6. Analytics, AI Insights, Predictions, and Anomalies tabs open their corresponding views.

Module Name: Health Monitoring Module

Screen No: 9

Screen Name: Vital detail

Description: The detail view provides the latest value, its recorded time, status, and available history or source context.

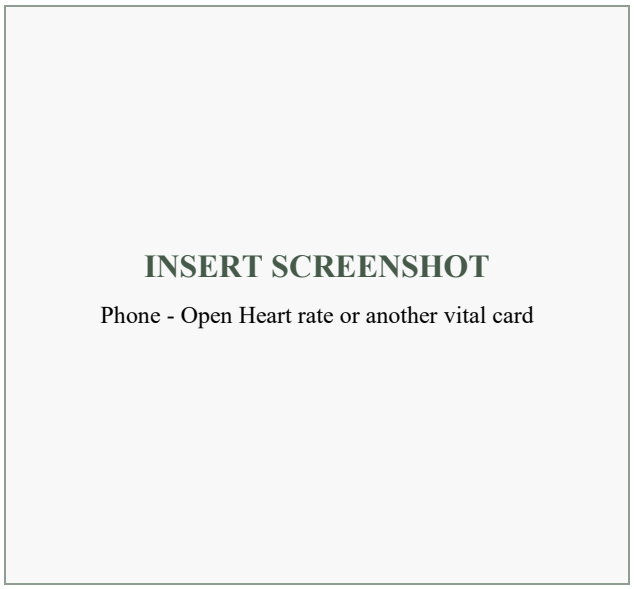


Figure 9: **Vital detail**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Metric title	Text	Yes	VARCHAR	N/A
Latest value	Reading	When recorded	NUMERIC	N/A
Unit	Text	When recorded	VARCHAR	N/A
Status label	Text	When recorded	VARCHAR	N/A
Measured time	Text	When recorded	TIMESTAMP	N/A
Reading history	List	When saved	OBJECT	N/A
Source context	Text	When available	VARCHAR	N/A
Close	Button	Yes	Action	N/A

Logic:

1. The caregiver taps a metric card from Home or Health.
    1.1. The detail view receives that metric and the selected patient context.
2. The app displays the latest available value, unit, status, and recorded time.
    2.1. If a live value is absent, it must not claim that an old saved value was just measured.
3. The app loads available saved history for that metric.
    3.1. If history exists, its real readings and times are shown.
    3.2. If history is empty, the view explains that no record is available.
4. Source or freshness context is shown when available.
5. The caregiver closes the detail view to return to the prior screen.

Module Name: Health Analytics Module

Screen No: 10

Screen Name: Monitoring analytics

Description: Caregivers can view measured trends and records within a chosen date range.

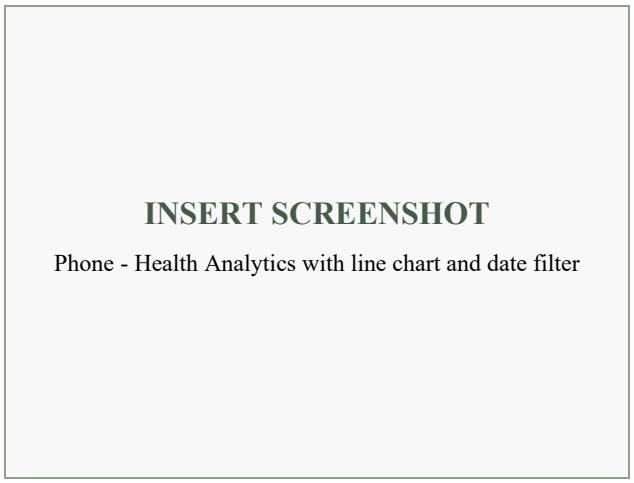


Figure 10: **Monitoring analytics**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Metric selector	Control	Yes	ENUM	N/A
From date	Date picker	Yes	DATE	N/A
To date	Date picker	Yes	DATE	N/A
Refresh	Button	Yes	Action	N/A
Line chart	Chart mode	No	Action	N/A
Bar chart	Chart mode	No	Action	N/A
Plotted reading	Chart point	When saved	NUMERIC	N/A
Reading time	Text	When saved	TIMESTAMP	N/A
Minimum	Statistic	When saved	NUMERIC	N/A
Maximum	Statistic	When saved	NUMERIC	N/A
Mean or change	Statistic	When saved	NUMERIC	N/A
Records list	List	When saved	OBJECT	N/A

Logic:

1. The caregiver selects a metric and chooses From and To dates.
    1.1. The app keeps date selection in the user's local calendar context.
2. The caregiver taps Refresh to load saved readings for that patient and range.
    2.1. If loading fails, an error is shown and no fabricated chart is drawn.
    2.2. If no readings match, the view explains that the range is empty.
3. If readings exist, the app plots actual values and displays their original timestamps.
    3.1. Minimum, maximum, mean, and change are calculated from the displayed records.
    3.2. For daily steps, one saved total per day is used rather than summing cumulative samples.
4. The caregiver switches between Line and Bar without changing the source records.
5. Tapping a plotted point shows its value and recorded date and time.
6. The caregiver may reverse the record list order or show more records when available.

Module Name: Health Analytics Module

Screen No: 11

Screen Name: Insights and anomalies

Description: The app presents general wellness context and flags readings that match its review rules.

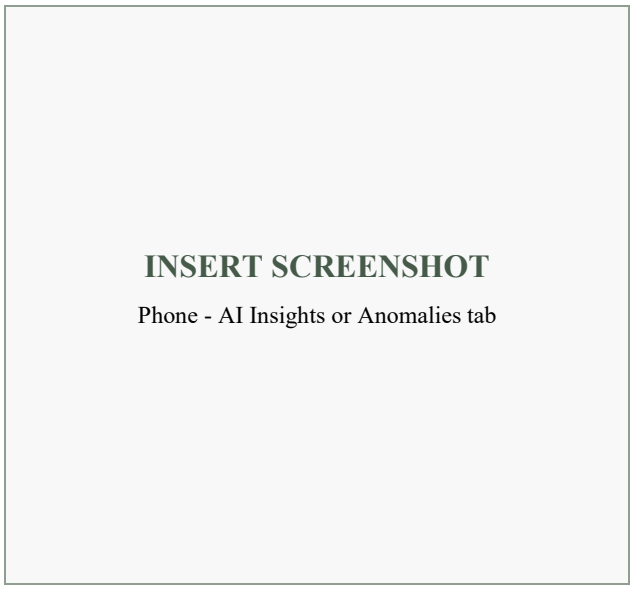


Figure 11: **Insights and anomalies**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
AI Insights tab	Tab	Yes	Action	N/A
Predictions tab	Tab	Yes	Action	N/A
Anomalies tab	Tab	Yes	Action	N/A
Metric name	Text	When analyzed	VARCHAR	N/A
Latest recorded value	Text	When saved	NUMERIC	N/A
Status label	Text	When analyzed	VARCHAR	N/A
Recorded time	Text	When saved	TIMESTAMP	N/A
General guidance	Text	When analyzed	TEXT	N/A
Insufficient data state	Text	When needed	VARCHAR	N/A

Logic:

1. The caregiver opens AI Insights, Predictions, or Anomalies under Health.
2. The app reads available patient measurements and their recorded times.
    2.1. If a metric has no usable reading, the view shows a waiting or missing-data state.
    2.2. If a prediction lacks enough distinct readings, the prediction remains pending.
3. With sufficient records, the app displays a trend, status, or anomaly derived from those records.
    3.1. A concerning measurement is presented as an app review finding, not a medical diagnosis.
    3.2. Experimental estimates are identified as such.
4. General wellness guidance may suggest confirming a concerning reading or contacting a professional.
5. The caregiver can return to Vitals or Analytics to inspect the underlying recorded values.

Module Name: Health History Module

Screen No: 12

Screen Name: Health records and sleep sync

Description: Recorded health history, checkups, and sleep import are shown for the selected older adult.

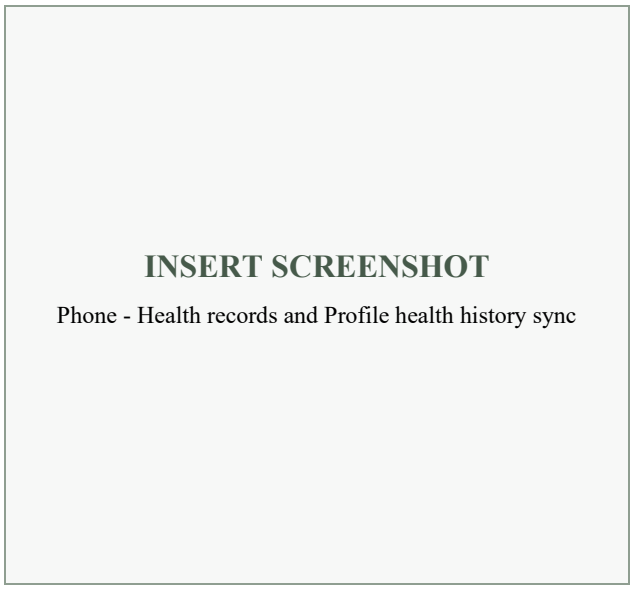


Figure 12: **Health records and sleep sync**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Health records entry	Button	Yes	Action	N/A
Saved checkups	List	When saved	OBJECT	N/A
Visit date	Text	When saved	DATE	N/A
Doctor	Text	When saved	VARCHAR	N/A
Medical notes	Text	When saved	TEXT	N/A
Sleep history	List	When imported	OBJECT	N/A
Sync health history	Button	No	Action	N/A
Sync status	Text	Yes	VARCHAR	N/A
Empty or error state	Text	When needed	VARCHAR	N/A

Logic:

1. The caregiver opens Health records or the Health History area for the selected patient.
    1.1. Saved checkups and history are loaded from that patient's records.
2. The caregiver may tap Sync health history to request available sleep records.
    2.1. The phone must have the required Samsung Health connection and patient watch association.
    2.2. If authorization or upload fails, the app shows status feedback without deleting saved history.
3. If a new sleep session is available, it is imported under the selected patient with its date and time.
    3.1. A repeated sync does not need to create a second display entry for the same saved session.
    3.2. If no new session exists, the app distinguishes that result from having no saved records at all.
4. The caregiver can inspect existing checkups and edit permitted records through the available form.

Module Name: Location and Safety Module

Screen No: 13

Screen Name: Live location and safe zone

Description: The map displays the selected watch route and a caregiver-set safe zone.

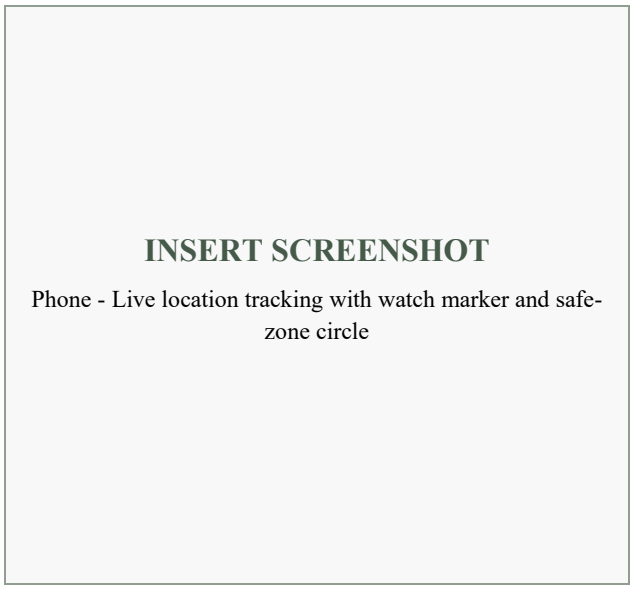


Figure 13: **Live location and safe zone**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Live map	Map	Yes	OBJECT	N/A
Watch marker	Map marker	When fixed	COORDINATE	N/A
Recent route	Polyline	When saved	OBJECT	N/A
Location source	Text	When available	VARCHAR	N/A
Fix age	Text	When fixed	DURATION	N/A
Accuracy	Text	When fixed	NUMERIC	meters
Safe-zone center	Map marker	When set	COORDINATE	N/A
Safe-zone radius	Control	When set	INTEGER	meters
Follow location	Button	No	Action	N/A
Waiting or stale status	Text	When needed	VARCHAR	N/A

Logic:

1. The caregiver opens Live location tracking for the selected patient and paired watch.
2. The app checks for an accepted watch position.
    2.1. If none exists, it shows waiting or stale status instead of using the caregiver phone as the patient marker.
    2.2. If one exists, it shows the marker with reported source, age, and accuracy when available.
3. The caregiver places the safe-zone center on the intended home location and chooses a radius.
    3.1. The zone is stored for the selected patient, not for another account.
4. As accepted watch positions arrive, the marker and recent route update.
    4.1. If the caregiver pans away, Follow location returns the map to the latest marker.
    4.2. If the watch loses internet or location, the map retains context but indicates the age of the last fix.
5. A safe-zone crossing should be interpreted with GPS accuracy in mind before the caregiver acts.

Module Name: Care Plan Module

Screen No: 14

Screen Name: Medications

Description: Caregivers can view scheduled medicines, dose status, image, and actions.



Figure 14: **Medications**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Medication image	Image	When saved	FILE	N/A
Medication name	Text	Yes	VARCHAR	N/A
Dosage	Text	When saved	VARCHAR	N/A
Frequency	Text	When saved	ENUM	N/A
Reminder time	Text	If scheduled	TIME	12-hour
Start date	Text	When saved	DATE	N/A
End date	Text	When saved	DATE	N/A
Instructions	Text	When saved	TEXT	N/A
Dose status	Badge	Yes	ENUM	N/A
Taken	Button	Yes	Action	N/A
Edit	Button	Yes	Action	N/A
Delete	Button	Yes	Action	N/A
Add medication	Button	Yes	Action	N/A

Logic:

1. The caregiver opens Care plan and selects Medications for the current patient.
2. The app loads active medication schedules and each patient's recorded dose status.
    2.1. Each card shows its own image or a placeholder, name, dosage, frequency, and instructions when present.
    2.2. Scheduled medicines show reminder times in a readable 12-hour format.
3. The caregiver taps Add or Edit to enter or correct a medication.
    3.1. If required details or times are invalid, the editor explains the issue without discarding the draft.
    3.2. If saving succeeds, the card updates and applicable reminders are prepared.
4. The caregiver taps Taken after a dose is actually taken.
    4.1. The app saves that confirmation and refreshes the displayed dose status.
5. If Delete is selected, the app asks for confirmation before deactivating the schedule.
    5.1. Applicable reminders are cancelled after the schedule is removed.
6. If no medications exist, the app shows its empty-state prompt.

Module Name: Care Plan Module

Screen No: 15

Screen Name: Prescription review

Description: An uploaded prescription produces an editable draft that the caregiver checks against the original before saving.

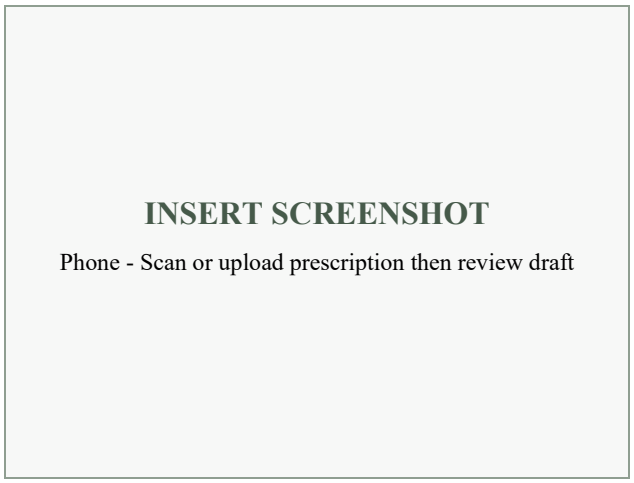


Figure 15: **Prescription review**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Elle illustration	Image	Yes	FILE	N/A
Take photo	Button	No	Action	N/A
Select image	Button	No	Action	N/A
Prescription preview	Image	When selected	FILE	N/A
Extract editable drafts	Button	When selected	Action	N/A
Original prescription	Image link	When extracted	FILE	N/A
Medication name	Textbox	For save	VARCHAR	N/A
Dosage	Textbox	For save	VARCHAR	N/A
Frequency	Choices	For save	ENUM	N/A
Reminder times	Time controls	If scheduled	TIME	N/A
Start date	Date control	For save	DATE	N/A
End date	Date control	No	DATE	N/A
Instructions	Textbox	No	TEXT	N/A
Prescribing doctor	Textbox	No	VARCHAR	N/A
Confirmation	Checkbox	For draft save	BOOLEAN	N/A
Confirm and save	Button	For draft save	Action	N/A

Logic:

1. The caregiver opens the optional prescription import from Care plan.
2. The caregiver takes a clear photo or selects an existing prescription image.
    2.1. Without a selected image, extraction remains unavailable.
3. The app extracts readable information into an editable draft.
    3.1. If text is unclear or a field is missing, that value stays blank or is marked for review.
    3.2. Extraction never activates a schedule by itself.
4. The caregiver opens the original image and compares every draft field with it.
    4.1. Name, dosage, frequency, times, dates, instructions, and prescriber can be corrected.
    4.2. If the original is unclear, the caregiver can use manual entry or consult the prescriber.
5. The caregiver checks the confirmation only after reviewing the original prescription.
    5.1. Without confirmation, Confirm and save medication remains blocked.
    5.2. With confirmation and valid fields, the app saves the medication and applicable schedule.
6. The saved medication appears in Care plan with its reminders when scheduled.

Module Name: Care Plan Module

Screen No: 16

Screen Name: Appointments and schedule

Description: Appointments and care tasks are displayed in time order with upcoming and completed states.



Figure 16: **Appointments and schedule**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Appointments tab	Tab	Yes	Action	N/A
Appointment title	Textbox	For new visit	VARCHAR	N/A
Doctor name	Textbox	No	VARCHAR	N/A
Appointment date	Date picker	For new visit	DATE	N/A
Appointment time	Time picker	For new visit	TIME	N/A
Location	Textbox	No	VARCHAR	N/A
Notes	Multiline textbox	No	TEXT	N/A
Save	Button	For new visit	Action	N/A
Complete	Button	For active visit	Action	N/A
Edit	Button	When saved	Action	N/A
Delete	Button	When saved	Action	N/A
Schedule tab	Tab	Yes	Action	N/A
Activity phase	Text	When saved	ENUM	N/A

Logic:

1. The caregiver opens Care plan and selects Appointments.
2. The caregiver taps Add and enters a title, appointment date, and appointment time.
    2.1. Doctor name, location, and notes can be added when known.
3. The caregiver taps Save.
    3.1. If a required value or date is invalid, the form remains open with feedback.
    3.2. If the database save fails, the entered draft remains available for retry.
    3.3. If saving succeeds, the new appointment appears in the list.
4. When notification permission and delivery are available, an appointment reminder is prepared for the saved time.
5. A visit may be edited, marked Complete, or deleted through its visible actions.
    5.1. Changes to the visit update its displayed date and time and applicable reminder.
6. The Schedule tab orders upcoming doses and visits by actual date and time and distinguishes completed activity.

Module Name: Alerts and Emergency Module

Screen No: 17

Screen Name: Alerts and SOS

Description: Caregivers review vital alerts and SOS requests and can contact priority people.

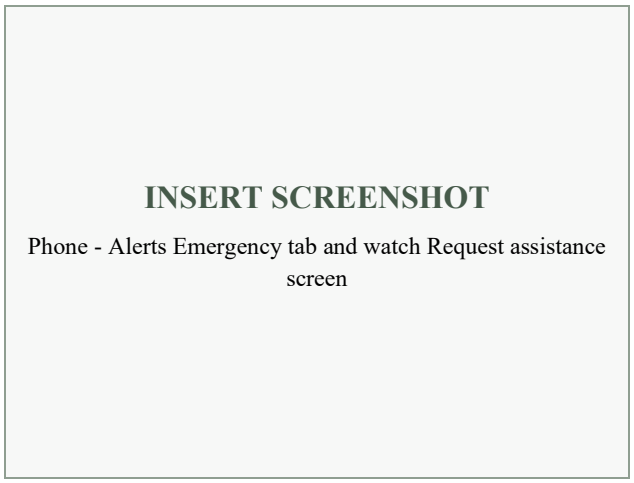


Figure 17: **Alerts and SOS**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Alerts tab	Tab	Yes	Action	N/A
Notifications tab	Tab	Yes	Action	N/A
Emergency tab	Tab	Yes	Action	N/A
Alert card	Card	When saved	OBJECT	N/A
Severity	Badge	When saved	ENUM	N/A
Alert time	Text	When saved	TIMESTAMP	N/A
SOS button	Button	Yes	Action	N/A
Watch Send SOS	Watch button	Yes	Action	N/A
Caregiver contact	Contact	When saved	VARCHAR	N/A
Family contact	Contact	When saved	VARCHAR	N/A
Doctor contact	Contact	When saved	VARCHAR	N/A
Review and respond	Button	When saved	Action	N/A
Enable background notifications	Button	No	Action	N/A

Logic:

1. The caregiver opens Alerts to review health warnings, notifications, and emergency history.
2. A wearer who needs help opens SOS on the watch.
    2.1. Opening the watch screen does not send an SOS; the wearer must tap Send SOS.
    2.2. If the watch has no pairing or internet, it reports the problem and the wearer should contact help directly.
3. On the phone, tapping SOS also requires explicit confirmation.
    3.1. After a successful request, the event appears in the selected patient's emergency history.
    3.2. If the request fails, the app shows an error instead of claiming delivery.
4. With configured push, allowed notifications, and internet, the caregiver phone may receive a background alert.
5. The contact list shows caregiver first, family second, and doctor third when numbers are saved.
    5.1. The caregiver chooses a contact; the app opens a supported dial action rather than placing a call silently.
6. The caregiver opens Review and respond to record an action or resolve the saved event.

Module Name: Elle Assistant Module

Screen No: 18

Screen Name: Elle care assistant

Description: Elle summarizes available care data and provides navigation to related app screens.

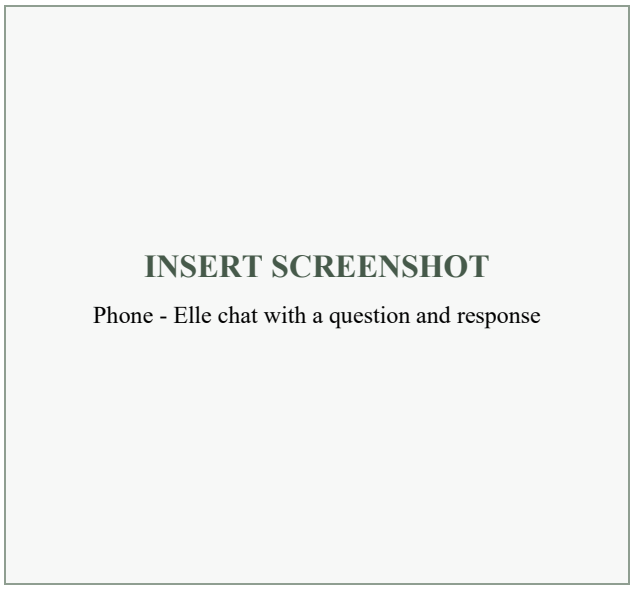


Figure 18: **Elle care assistant**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Elle avatar	Image	Yes	FILE	N/A
Quick topic chips	Buttons	No	Action	N/A
Question input	Textbox	Yes	TEXT	N/A
Send	Button	Yes	Action	N/A
Thinking status	Indicator	While loading	BOOLEAN	N/A
User message	Message	When sent	TEXT	N/A
Elle reply	Message	When received	TEXT	N/A
Evidence reading	Context	When available	OBJECT	N/A
Related action	Button	When available	Action	N/A
New Chat	Button	No	Action	N/A

Logic:

1. The caregiver opens Elle for the selected patient context.
2. The caregiver types a question or taps a quick topic.
    2.1. Empty input is ignored and no assistant request is sent.
3. While the request is pending, Elle shows a thinking state.
    3.1. If the service fails, a readable error appears and no false answer is presented.
    3.2. If a reply succeeds, the question and answer appear in the conversation.
4. When a response uses saved patient data, available evidence appears with the reply.
    4.1. Missing health data is not replaced with invented measurements.
5. If the reply offers a related action, tapping it opens the relevant care screen.
6. Tapping New Chat clears the current view and starts a fresh conversation session.

Module Name: Settings and Reports Module

Screen No: 19

Screen Name: Profile and exports

Description: Settings contain account and patient information, integrations, emergency contacts, and PDF export actions.

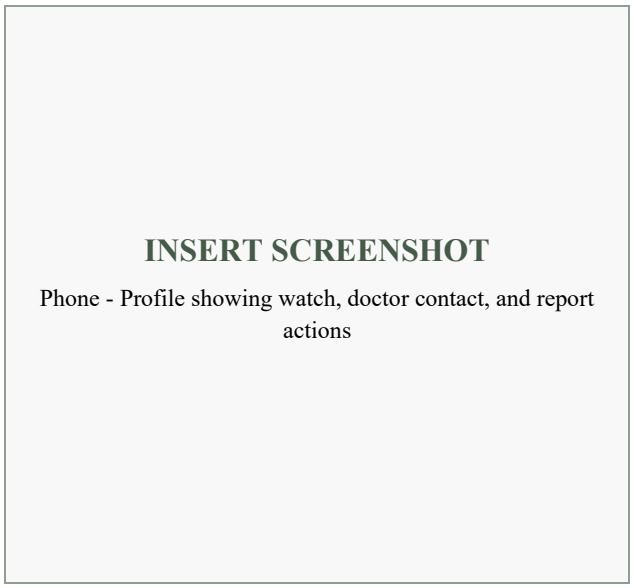


Figure 19: **Profile and exports**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Caregiver name	Text	Yes	VARCHAR	N/A
Account email	Text	Yes	VARCHAR	N/A
Selected patient	Text	When saved	VARCHAR	N/A
Galaxy Watch6 status	Status	When paired	ENUM	N/A
Health history sync	Button	No	Action	N/A
Doctor contact	Card	When saved	VARCHAR	N/A
Emergency contacts	Cards	When saved	VARCHAR	N/A
Health vitals PDF	Button	No	Action	N/A
Medication schedule PDF	Button	No	Action	N/A
Sign out	Button	Yes	Action	N/A

Logic:

1. Profile opens with the current caregiver account and selected older adult.
2. The app shows account identity and available Galaxy Watch6 connection status.
    2.1. Contact cards show saved doctor and emergency information when present.
3. The caregiver may edit a contact or patient detail through the corresponding form.
    3.1. Invalid fields or a failed save keep the form available for correction.
4. The caregiver may tap Sync health history for available recorded sleep data.
    4.1. A sync error displays a status message while existing saved records remain available.
5. The caregiver selects Health vitals PDF or Medication schedule PDF.
    5.1. If the selected patient or required records are missing, the app shows guidance.
    5.2. If records exist, a patient-scoped PDF is generated and the share sheet opens.
6. Sign out ends the active account session before another caregiver signs in.

Module Name: Administration Module

Screen No: 20

Screen Name: Admin sign in and overview

Description: An authorized administrator signs in to review high-level account and health-activity information.

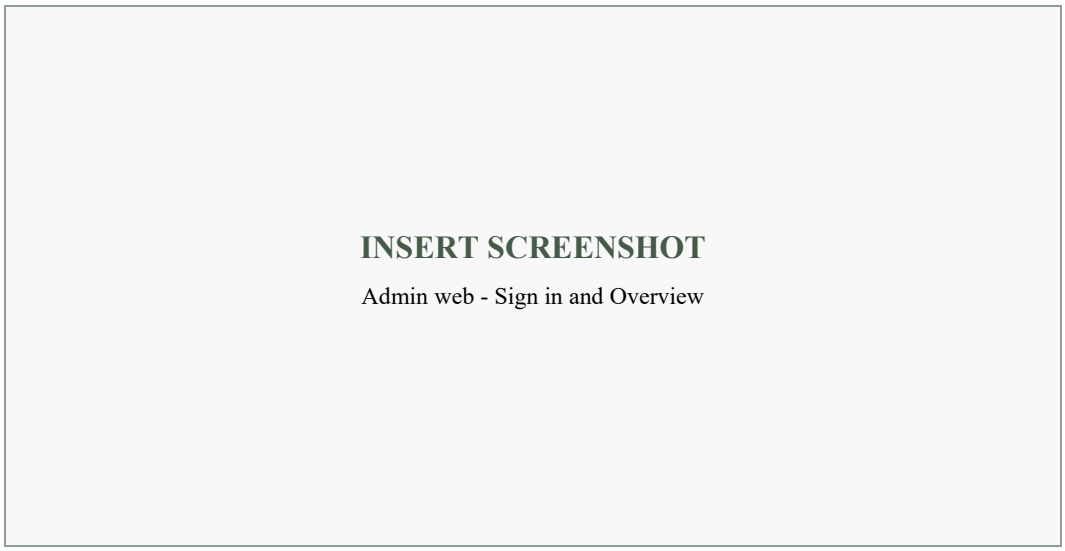


Figure 20: **Admin sign in and overview**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Username or email	Textbox	Yes	VARCHAR	254
Password	Password textbox	Yes	VARCHAR	N/A
Sign in	Button	Yes	Action	N/A
Forgot password	Button	No	Action	N/A
Login message	Text	When needed	VARCHAR	N/A
Caregiver count	Metric	After sign-in	INTEGER	N/A
Older adult count	Metric	After sign-in	INTEGER	N/A
Connected device count	Metric	After sign-in	INTEGER	N/A
Suspended account count	Metric	After sign-in	INTEGER	N/A
Recent alerts	List	When saved	OBJECT	N/A
Recent emergencies	List	When saved	OBJECT	N/A

Logic:

1. The administrator opens the web portal and enters username or email and password.
2. The administrator submits Sign in.
    2.1. If credentials are invalid or the account lacks administrator access, the portal shows an error and stays at sign-in.
    2.2. If authorization succeeds, the portal opens Overview.
3. Forgot password opens the supported recovery form when needed.
    3.1. The administrator follows the code verification and password-reset steps shown there.
4. Overview requests caregiver, older-adult, connected-device, and suspended-account counts.
    4.1. If an API request fails, a visible status is shown rather than an invented count.
5. Recent alerts and emergencies appear when saved data is available.
6. The sidebar opens Accounts, System monitor, Alerts and notifications, Reports, or Security.
7. Sign out ends the administrator session.

Module Name: Administration Module

Screen No: 21

Screen Name: Admin accounts and monitor

Description: The administrator reviews caregiver and older-adult records and recent saved syncs.

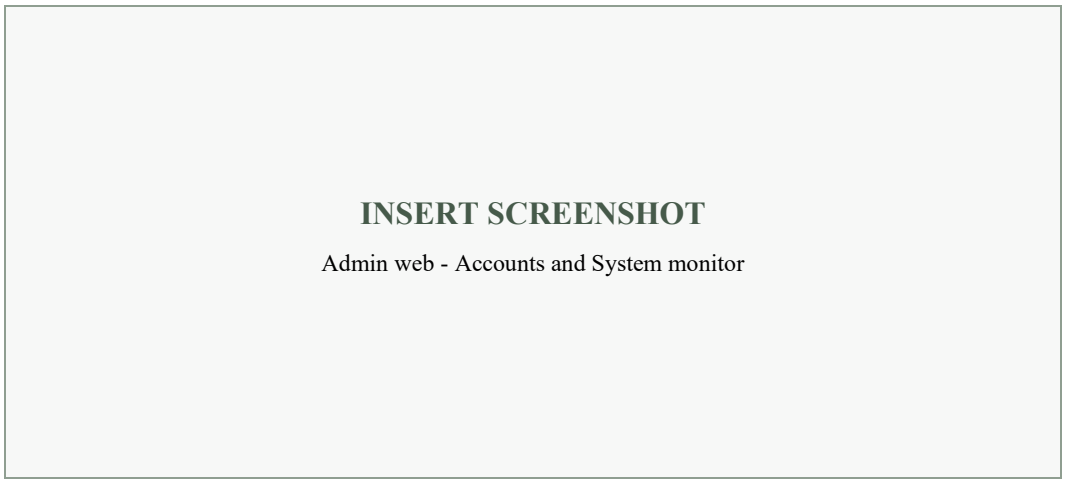


Figure 21: **Admin accounts and monitor**

ITEMS	TYPE	REQUIRED?	DATA TYPE	SIZE
Caregiver accounts	Table	When saved	OBJECT	N/A
Caregiver contact	Table field	When saved	VARCHAR	N/A
Caregiver role	Table field	When saved	ENUM	N/A
Caregiver status	Table field	When saved	ENUM	N/A
Linked older adults	Table	When selected	OBJECT	N/A
Older adult age	Table field	When saved	INTEGER	N/A
Emergency contact	Table field	When saved	VARCHAR	N/A
Connected devices	Table	When saved	OBJECT	N/A
Paired device count	Table field	When saved	INTEGER	N/A
Recent health syncs	Table	When saved	OBJECT	N/A
Saved at	Table field	When saved	TIMESTAMP	N/A
Measured at	Table field	When saved	TIMESTAMP	N/A
Refresh	Button	No	Action	N/A

Logic:

1. The administrator opens Accounts and loads the current page of caregiver records.
    1.1. If no records exist, the table shows its empty state.
2. Selecting a caregiver loads the older adults linked to that caregiver.
    2.1. The administrator can open permitted edit forms for caregiver or patient information.
    2.2. If a field is invalid or saving fails, the form shows feedback without claiming an update.
3. The administrator opens System monitor to review connected-device state and recent saved health syncs.
    3.1. Saved at is the database write time; Measured at is the reading time.
    3.2. The device table reflects reported onboarding connection state, not live sensor telemetry.
4. Refresh requests updated administration data.
    4.1. If the API fails, show a visible error while retaining the current page context.
5. Reports, Alerts, and Security provide their own administrative actions through the sidebar.

Screenshot Capture Checklist

Replace each gray figure box with the screen named below. Use the same patient account throughout a demonstration. Hide personal names, phone numbers, medical notes, tokens, and exact home coordinates before sharing the document.

1. Figure 1. Phone - Sign in, showing Google button
2. Figure 2. Phone - Tell us about you form
3. Figure 3. Phone - Who are you caring for form
4. Figure 4. Phone - Connect Galaxy Watch6 with pairing code
5. Figure 5. Watch - Main screen showing Start monitoring and readings
6. Figure 6. Watch - Blood oxygen measurement screen
7. Figure 7. Phone - Home dashboard with patient name and vital cards
8. Figure 8. Phone - Health Vitals tab with six metric cards
9. Figure 9. Phone - Open Heart rate or another vital card
10. Figure 10. Phone - Health Analytics with line chart and date filter
11. Figure 11. Phone - AI Insights or Anomalies tab
12. Figure 12. Phone - Health records and Profile health history sync
13. Figure 13. Phone - Live location tracking with watch marker and safe-zone circle
14. Figure 14. Phone - Care plan Medications tab with a saved medication card
15. Figure 15. Phone - Scan or upload prescription then review draft
16. Figure 16. Phone - Care plan Appointments and Schedule tabs
17. Figure 17. Phone - Alerts Emergency tab and watch Request assistance screen
18. Figure 18. Phone - Elle chat with a question and response
19. Figure 19. Phone - Profile showing watch, doctor contact, and report actions
20. Figure 20. Admin web - Sign in and Overview
21. Figure 21. Admin web - Accounts and System monitor